

CONCEPT NOTE: Ensuring effective guardrails and pilots for catalytic HFC destruction projects in Article 5 countries

February 2026

Context

Thanks in large part to the foundational work of the super pollutant ecosystem over many decades, there has been a recent surge of interest in near-term warming and super pollution mitigation, especially within the private sector. An influx of capital into super pollutant mitigation is expected from both companies (*via voluntary carbon credits and/or insetting projects*) and governmental buyers (*via Article 6 projects*). In order to direct and maximize the long-term impact of this push and to maintain trust, clear guardrails and enabling pre-conditions will be critical — ensuring that individual projects can succeed and the super pollutant community is driving in a clear and consistent direction towards scale-up and policy and industry support.

At the [Yale Clean Cooling Conference](#), **Cascade and the Carbon Containment Lab will be bringing together a small, invite-only expert group to discuss what a practical, testable set of data collection priorities and real-world guardrails should look like for an initial wave of catalytic VCM-funded LRM projects.** Attendees will discuss how to best pilot promising approaches to mitigate perverse incentives and measurement risks, with the goal of enabling the dramatic scale-up of these approaches while protecting integrity and avoiding fraud. While this workshop will largely focus on HFC destruction, ODS destruction will also be discussed.

Background

About [Cascade Climate](#)

Cascade is a philanthropic backed non-profit focused on propelling high-potential climate solutions from the margins to the mainstream. Our super pollution mitigation efforts aim to build a bridge from a handful of early, catalytically-funded demonstration projects toward scale-up through policies and market mechanisms that support embedding super pollutant reductions within industry value chains and national and subnational policies. Refrigerant management and transition represents our initial priority area within super pollutant mitigation.

About [Carbon Containment Lab](#)

The Carbon Containment Lab is a public charity advancing research and education for a public benefit. The CC Lab applies scientific, entrepreneurial, and investment expertise, creating an enabling environment for emergent climate solutions. The CC Lab's cooling program aims to bring confidence to the market for life cycle refrigerant management and alternative cooling technologies.

The Opportunity

Lifecycle refrigerant management (LRM) represents a rare convergence: a climate intervention that is both cost-effective and deployment-ready, while also strengthening supply chain resilience during a complex industrial transition — i.e., the HVAC and refrigeration sectors' transition towards lower-GWP refrigerants. The Montreal Protocol's Technology and Economic Assessment Panel (TEAP) [estimates](#) that between 2025 and 2050, the implementation of LRM best practices globally could prevent 39 gigatonnes of CO₂e — equivalent to roughly a year's worth of global energy sector emissions.

Building on the foundational work of leading climate organizations and entrepreneurial project developers, **we have identified “Article 5” (A5) countries under the Montreal Protocol as a high opportunity region for catalytic LRM work.** Initial LRM activities financed by the voluntary carbon market (VCM) can help to stand up refrigerant recovery capacity in areas with high refrigerant venting rates — setting the stage for Article 6 integration and eventual policy and industry uptake, including via regulation or other approaches like extended producer responsibility. Refrigerant recovery is a key bottleneck in A5 countries where gas venting rates are extremely high. In tandem with policies supporting the transition to low-GWP refrigerants and reclamation and reuse of gases, carbon market financing for recollect-to-destroy projects can support initial recovery capacity build-up in a highly additional way.

VCM refrigerant destruction projects — for both ODS and HFCs — have the potential to provide a quick path to successful projects and impact. However, while some methodologies for A5 contexts exist, they have yet to be fully tested across a wide range of contexts and markets. Effective market guardrails necessary for scale — especially for recently published or in-progress HFC destruction methodologies and projects — remain theoretical or under-tested. Real-world demonstration projects in A5 countries are a crucial way to test and determine what data can actually be collected; how chain-of-custody and verification requirements can be implemented; and whether theoretical guardrails could truly prevent perverse incentives in practice.

Early, innovative projects and initial methodologies have put forward thoughtful approaches for how to mitigate these potential risks - but determining how effective these mitigation approaches will be can only be resolved via demonstration projects and learning-by-doing on the ground, alongside careful monitoring of outcomes. The community needs to get these guardrails and pre-conditions right — and test them — if they want to benefit from the recent influx of interest into super pollutant mitigation from the private sector. If successful, these projects can accelerate the LRM transition in A5 countries and build the way for more integrated approaches to meet the needs of a warming world.

This workshop — **which is an invitation-only, pre-conference workshop to the Yale Cooling Conference** — will bring together key stakeholders to discuss and refine plans to meet these

needs. Focusing on methodology writers, project developers, raters and registries, and refrigerant experts, we aim to coalesce an open and informed discussion about the key risks, guardrails, data needs and testing approaches that could be deployed in the coming 1-2 years.

Logistics

Date: Thursday March 5th

Time: 12:30-2:30PM EST

- Light lunch of sandwiches provided at noon EST

Location: Yale School of Management, New Haven, CT

- In-person only under Chatham House rules.

Co-hosts: Cascade Climate and Carbon Containment Lab

- Cascade and CCLab will serve as the hosts and facilitators, but we hope that attending organizations will take an active and engaged role in the discussion.

Participants:

- ~20 attendees from across carbon registries, project developers, market supporting organizations (e.g., credit ratings agencies and HFC/VCM market experts), civil society groups, and financiers.
- Cascade leads: Brad Rochlin, Alex Dolginow
- CCLab leads: Anastasia O'Rourke, Sinead Crotty

Pre-Reads:

- To tee up discussion and make sure participants are well informed, Cascade and the CC Lab will provide several pre-reads for workshop participants — including a “*State of the voluntary, compliance and article carbon markets today*” for LRM-related registries, methodologies and projects, as well as some reflections on the key risks and guardrails that are already known based on recent research and experience to-date.

Workshop structure

Working tagline: Convene the actors who will be critical in driving forward early LRM carbon projects in Article 5 countries to discuss what a practical, testable set of data priorities and real-world guardrails should look like for initial VCM-funded HFC destruction projects, and how to design the right testing window to effectively evaluate perverse-incentive risk and measurement feasibility across priority pathways and countries.

Potential (WIP) goals and outcomes for this workshop:

- Clarify what data and monitoring is needed to prove HFC destruction projects are high-quality and avoid perverse incentives, in order to catalyze crowding-in of demand.
 - Identify data collection roles and responsibilities (e.g., *as projects come online, who is responsible for what data, and where is it coming from?*), which can be refined and shared publicly post-workshop.

- Facilitated discussion on practical-vs-ideal data collection requirements - what is possible today, and does that effectively mitigate potential project risks (*e.g., collecting in-country recovery rates or data needed to prove that virgin gas production isn't increasing as a result of HFC destruction crediting*).
- Develop a prioritized short-list of the most critical risk-mitigation approaches to trial in initial pilots, to supplement methodology requirements.

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- *Stretch goal:* Discussion on field-wide pilot timing and how to make maximal progress in a short experimental project window (*e.g., what is the right length of time, priority metrics and who is responsible for collecting, independent review requirements, etc.*).
 - *Stretch goal:* Expert perspectives on an initial project category “risk-tiering” framework that maps project types against risks or both gas type (*e.g., ODS vs. multi-gas HFC vs single-gas HFC*) and gas sources (*e.g., EOL vs. servicing, stockpiled vs. non-stockpiled*), and which areas require additional data & controls within or outside of methodological requirements.